**Lecture III – Decision Tree Classification**

**Decision Trees:**

Decision trees are used to identify labels for objects based on data attributes. It is useful to find which attributes are most useful. Each attribute is considered iteratively to partition the dataset into subgroups. The split that results in the greatest information gain is preferred.

**Entropy:**

$$E = -\sum_{x=1}^{n} p_x \bullet \log_2 p_x$$

Example, entropy for a population with two classes could be written like this:

$$E = -(p1 \log_2 p1 + p2 \log_2 p2)$$

Where p1 represents the ratio of how many objects are in that class divided by the total number of objects. E.g., in a population with 3 stars and 2 diamonds, p1 would equal 3/5, and p2 would equal 2/5.

**Information Gain**:

Information Gain is the difference between the entropy of the parent minus the weighted entropy of the child nodes.

**Entropy of Parent – (Entropy of child A X fraction in child A + Entropy of Child B X fraction in child B)**

For example, if the entropy of the parent is .811, the entropy for the 10 objects in child A is .4689, and the entropy for the 2 objects in child B is 0, then:

**Δ = .811 – (.4689 X 10/12 + 0 X 2/12)**

**Δ = .4203**

The process repeats to further split the subgroups into more subgroups until entropy reaches zero or an acceptable threshold.

The uniqueness of an attribute can affect the outcome of the splitting process. Highly unique attributes might find classifications with fewer steps, but not prove accurate when applied to other datasets. For example, we could quickly classify a student's major by splitting on a student ID field. But we wouldn't be able to use that decision tree to predict the major of a student whose student ID was not used in the construction of the decision tree.

To decrease the uniqueness of an attribute, we can employ discretization methods discussed earlier. For continuous attributes, the values are sorted. Then the midpoint between each value is found. The algorithm then splits on each midpoint and the split with the highest information gain is selected.

**Bibliography**

Guess Who Example

Tan, Steinbach, Karpatne, and Kumar. Introduction to Data Mining, second edition. Pearson, Hudson, NY, 2019.

Han, Kamber, Pei. Data Mining Concepts and Techniques, second edition. University of Illinois at Urbana-Champaign. Elsevier, San Francisco, CA, 2006.